

Sarthak Mishra

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Technical Skills

Languages: Java, Python, C++, SQL, Go

Backend & Data: Spring Boot, REST APIs, Microservices, Kafka, Spark, Redis, PostgreSQL, MongoDB, Hibernate, Maven, DynamoDB

Cloud & Infrastructure: AWS (Lambda, SWF, S3, RDS, CloudFront, VPC, EC2), Docker, Kubernetes, Linux, CI/CD, ElasticSearch

Networking: TCP/IP, BGP, NETCONF, SDN, Control Plane, Data Plane, Network Automation

Core Systems: Distributed Systems, Concurrency, Fault Tolerance, Sharding, Caching, System Design

AI Tools: Claude Code, Cursor, Codex

Professional Experience

Amazon Web Services (AWS)

Seattle, WA

Network Development Engineer Intern – Internet Edge

May 2026 – Aug 2026

- Automated end-to-end network infrastructure deployment for **AWS Data Transfer Terminal** using **AWS SWF and Lambda**, reducing engineer deployment time from **2 weeks to under 1 hour** and delivering **\$1M+ in estimated annual engineering savings**.
- Optimized **control-plane and data-plane networking software** across AWS data centers and colocation sites, improving infrastructure readiness for **HPC and distributed AI/ML workloads**.
- Developed automation and validation workflows for production network infrastructure, reducing manual deployment operations and enabling repeatable execution across geographically distributed sites.

Nokia

Chennai, India

Software Engineer – Distributed Systems & Cloud Infrastructure

Jul 2021 – Dec 2024

- Developed **Java/Spring Boot microservices** for Nokia's Software-Defined Networking platform, enabling operators to visualize, validate, and provision fixed-network devices while reducing end-to-end provisioning time from **5 days to 55 minutes**.
- Designed a sharded device configuration **DataStore** on the Access Controller to distribute network configuration state, improving configuration update/write performance by **95%** under concurrent workloads.
- Designed and developed a microservice from scratch to retrieve and visualize device configuration and operational state through **NETCONF GET RPCs**, enabling on-demand inventory and customer-controlled network configuration snapshots.
- Upgraded the Kubernetes **Ingress Controller** and optimized deployment behavior across cluster nodes, reducing infrastructure pod startup time from **10 minutes to approximately 1 minute**.
- Re-architected legacy Kubernetes deployments from **Recreate to RollingUpdate**, enabling progressive service upgrades and **zero-downtime deployments** for customer environments.
- Increased service throughput by **30%** through JVM heap/GC tuning and root-cause analysis of multithreaded contention and race conditions using **jstack and thread dumps**.
- Engineered resilience mechanisms including circuit breakers, retries, and Kafka consumer/broker tuning to prevent cascading failures and maintain reliable event processing across distributed services.
- Represented Nokia during global customer escalations, performing real-time root-cause analysis across application, infrastructure, and networking layers to isolate performance bottlenecks, deployment failures, and compatibility issues.
- Built a concurrent configuration-validation framework capable of validating **6M+ XML nodes in under 15 seconds**, accelerating release validation and reducing deployment defects.

Arizona State University

Tempe, AZ

Software Engineer (Part-Time)

May 2025 – Present

- Developed backend services for **Smart-DART**, an Uber-like campus transportation platform for students with accessibility needs, automating enrollment and operational workflows to reduce student onboarding time by **90%**.
- Engineered asynchronous **REST API integrations** with ASU GIS services to dynamically ingest geographic stops and incorporate them into distance and path-calculation workflows, enabling dynamic point-to-point routing.
- Designed role-based application workflows for riders, drivers, and administrators, separating permissions and operational responsibilities across the transportation platform.

Technical Projects

Distributed Real-Time Streaming Pipeline

Fall 2024

Apache Kafka, Spark Structured Streaming, AWS S3, Redis, Python

- Architected a fault-tolerant streaming pipeline processing **1M+ events/day**, implementing exactly-once processing with **Spark checkpointing** and durable recovery semantics.
- Engineered low-latency enrichment through **Redis** and implemented backpressure, retries, and **Dead Letter Queues (DLQs)** to preserve availability and data integrity during downstream failures.

LLaMA-2 Edge Inference Optimization

Spring 2025

C++, Python, PyTorch, CUDA

- Optimized LLaMA-2 inference for resource-constrained environments using memory-efficient attention, **KV caching**, and weight quantization.
- Reduced inference latency and peak GPU memory utilization by **45%**, enabling inference on a single **12GB GPU** while preserving model output quality.

Distributed Key-Value Store

Fall 2026

Technologies: Java, gRPC, RocksDB, Docker, Consistent Hashing

- Designed a distributed key-value store supporting **horizontal partitioning, replication, and quorum-based reads/writes**, sustaining **1K+ operations/sec** across 10 nodes.
- Implemented **consistent hashing and automatic shard rebalancing**, redistributing partitions after node additions/failures while maintaining service availability.
- Engineered write-ahead logging, caching, and replica recovery, reducing **p99 read latency by 50%** and recovering failed nodes within 30 seconds.

Education

Arizona State University, Tempe, AZ

Master of Science in Computer Science

Jan 2025 – Dec 2026

GPA: 4.0/4.0

SRM Institute of Science and Technology, Chennai, India

Bachelor of Technology in Computer Science

Aug 2017 – May 2021

Grade: 91.03% (First Class with Distinction)

Certifications

- **Microsoft Certified:** Azure Data Engineer Associate; Azure AI Engineer Associate
- **Cloud:** AWS Solutions Architect – Associate; Google Cloud Architecting with Google Compute Engine
- **Systems:** Linux Command Line (CompTIA); Kubernetes Engine: Qwik Start